

Worksheet 6 - Z-scores

Monday, February 16, 2026

Math 58B - Jo Hardin

Name: _____

Names of people you worked with: _____

What is the best part about the rain?

Task:

The example below allows for a comparison between two athletes based on speed and strength. The following information is provided about the sample of individuals who were measured:

- Speed is measured by the time required to run a distance of 40 yards, with smaller times indicating more desirable (faster) speeds. From the data, the times to run 40 yards have a mean of 4.60 seconds and a standard deviation of 0.15 seconds.
- Strength is measured by the amount of weight lifted, with more weight indicating more desirable (greater) strength. From the data, the amount of weight lifted has a mean of 310 pounds and a standard deviation of 25 pounds.
- The following information is provided about the distribution of runs and lifts across all the players:

	mean	std dev
Time to run 40 yards	4.60 sec	0.15 sec
Amount lifted	310 lbs	25 lbs

1. Calculate Z scores for the following two players and each of the tasks (4 total Z scores):

	Player A	Player B
Time to run 40 yards	4.42 sec	4.57 sec
Amount lifted	370 lbs	375 lbs

2. Based only on the Z scores, and considering both athletic characteristics as equally valuable, which player would you rather choose for your team? Explain.

Solution:

$$\begin{aligned} Z_{Aspeed} &= \frac{4.42 - 4.6}{0.15} = -1.2 & Z_{Astrength} &= \frac{370 - 310}{25} = 2.4 \\ Z_{Bspeed} &= \frac{4.57 - 4.6}{0.15} = -0.2 & Z_{Bstrength} &= \frac{375 - 310}{25} = 2.6 \end{aligned}$$

After calculating Z scores, it is found that Player B is only slightly stronger than Player A, but Player A is considerably faster than Player B. Because the question advised us to consider both criteria as equally valuable, Player A is the better choice.